

Heuristic Search:-

↳ Heuristic is a technique that improves the efficiency of search process possibly by sacrificing claims of completeness.

↳ general purpose heuristic
↳ domain specific heuristic, you can create by exploiting the knowledge.

→ Travelling Salesman problem:-

- (i) Select any city as starting city.
- (ii) For selecting next city, look at all cities not yet visited and select the one closest to current city.
- (iii) Repeat step 2 untill all the cities are visited.

Time complexity :- $N^2 \leftarrow N!$

heuristic make $N!$ to N^2

Subject: Advantages of Heuristic Search

↳ (i) For real world problems

(ii) For unstructured information it is difficult to find optimality.

→ without heuristics, we would be hopelessly ensnared in a combinatorial exploration

→ rarely we need optimal solution

→ approx. may not be very good in the worst case

but worst cases rarely happens

→ Why a heuristic works?

↳ often leads to better understanding of the problem.

Heuristic f^n :- $f(x) \rightarrow y$

Heuristic f^n is a f^n that maps from problem state description to measures of desirability often represented as number

→ Which aspects of problem space are considered, how those aspects are evaluated and weights given to individual aspects?